

Date: June 1st, 2026

Course Instructor(s)

Dr. Shakil Ahmed, Mr. Usama Antuley, Dr. Fahad Sherwani,
Ms. Anam Hamid, Dr. Farrukh Salim Shaikh, Mr. Farooq Zaidi,
Ms. Abeeha Sattar, Ms. Alina Arshad, Ms. Javeria Farooq, Mr. Sandesh Kumar

Final Exam

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 8

Roll No

Section

Student Signature

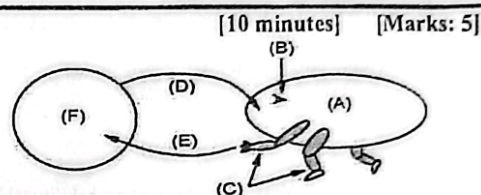
INSTRUCTION:

- Attempt all the questions in-order.
- Solve the problems as per duration of each question mentioned.

CLO #2: Express appreciation of the goals, subareas, achievements, and difficulties of AI

Question 01:

- a) What does each of the lettered items in the following diagram represent? (1.5 mark)



- b) Construct a complete PEAS description for the following task environment, then argue whether a simple reflex or utility-based agent is more appropriate and why. (1 mark)

An AI tutor that teaches high-school mathematics. It selects which problem to present next, explains concepts when the student is stuck, adapts difficulty based on performance, and must keep students engaged while maximizing learning outcomes.

- c) Classify the following environment along all five environment dimensions and justify each classification. (2.5 marks)

A self-driving car navigates an urban city during rush hour. Its cameras and radar give partial views of the road (other vehicles may be occluded). Weather is unpredictable. The car must decide moment-to-moment: brake, accelerate, or steer. Multiple autonomous vehicles from different companies operate on the same roads simultaneously, and each vehicle's actions affect road conditions for others.

- Classify and justify: Fully Observable vs. Partially Observable.
- Classify and justify: Deterministic vs. Stochastic.
- Classify and justify: Episodic vs. Sequential.
- Classify and justify: Static vs. Dynamic.
- Classify and justify: Discrete vs. Continuous.

Additionally, explain whether this is a single-agent or multi-agent environment and why that matters for algorithm choice.

CLO # 3: Apply methods of blind as well as informed search on the problem related to the programs.

Question 02: Part (a)

[15 minutes] [Marks: 7.5]

Consider assigning colors to a checkerboard so that squares that are adjacent vertically or horizontally do not have the same color. We know that this can be done with only two colors, say red (R) and black (B). We will limit our discussion to five squares on a 3x3 board, numbered as follows:

1	2	3
4	5	

Let's look at the CSP formulation of this problem. Let the squares be the variables and the colors be the values. All the variables have domains {R, B}.

- If we run full constraint propagation on the initial state, what are the resulting domains of the variables? (0.5 marks)
- Say, instead, the initial domain of variable 5 is restricted to {B}, with the other domains as before. If we now run full constraint propagation, what are the resulting domains of the variables? (2 marks)
- If in the initial state (all variables have domains {R, B}), we assign variable 1 to R and do forward checking, what are the resulting domains of the other variables? (2.5 marks)
- If we forward checking with dynamic variable ordering, using the most-constrained-variable strategy (Variable ordering), give a list of all the variable assignments attempted, in sequence. (2.5 marks)

Question 02: Part (b)

[15 minutes]

Four people, A, B, C, and D, are all looking to rent space in an apartment building. There are three floors in the building, 1, 2, and 3 (where 1 is the lowest and 3 is the highest). Each person must be assigned to some floor, but it's ok if more than one person is living on a floor. We have the following constraints:

- A and B must not live together on the same floor.
- If A and C live on the same floor, they must both be living on floor 2.
- If A and C live on different floors, one of them must be living on floor 3.
- D must not live on the same floor as anyone else.
- D must live on a higher floor than C.

Formulate this as a CSP, where each person has a variable and the variable values are floors.

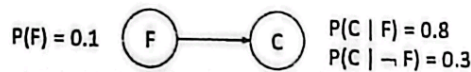
- (a) Draw the edges for the constraint graph representing this problem. Use binary constraints only. You do not need to label the edges. (3.5 marks)
- (b) Suppose we have assigned $C = 2$. Apply forward checking to the CSP with proper steps and proper constraint satisfaction steps. (4 marks)

CLO # 3: Apply methods of blind as well as informed search on the problem related to the programs

Question 03: Part (a)

[10 minutes] [Marks: 5]

Consider the following Bayesian network, where F = having the flu and C = coughing:



- i) Write down the joint probability table specified by the Bayesian network.

F	C	
T	T	
T	F	
F	T	
F	F	

- ii) Determine the probabilities for the following Bayesian network so that it specifies the same joint probabilities as the given one.



Question 03: Part (b)

[10 minutes] [Marks: 5]

3. A survey shows 56% of all American workers have a workplace retirement plan, 68% have health insurance, and 49% have both benefits. We select a worker at random.

- a) What is the probability he has neither health insurance nor a retirement plan? (1.5 marks)
- b) What is the probability he has health insurance, if he has retirement plan? (1.5 marks)
- c) Are having health insurance and a retirement plan independent event? (1 mark)
- d) Are these two benefits mutually exclusive? (1 mark)

CLO # 4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas

Question 04: Part (a)

[20 minutes] [Marks: 10]

Answer the following questions to the point

- i) Why do we need a train-test split? (1.5 marks)
- ii) Why do we sometimes need an additional validation dataset to do testing during model training? (1.5 marks)
- iii) Consider the table of flight delays below. (7 marks)

Flight ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Target	D	OT	OT	OT	D	D	OT	D	OT	OT	OT	OT	D	D	OT	OT
Predicted	OT	OT	OT	D	D	D	D	D	OT	OT	D	D	D	D	D	OT

D: Delayed Flight, OT: On Time flight

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every ID you see the target label of the flight being on time or delayed and the corresponding prediction of a machine learning model.

- Compute the corresponding confusion matrix. (2.5 marks)
- Calculate the accuracy, recall, precision and F1-measure corresponding to the confusion matrix. (2 marks)
- Considering the metrics you just calculated, fill in the following blanks with one word each. (2.5 marks)

If the model predicts a positive class, it is likely to be _____. However, for classifying positive instances correctly, the model performs similarly to _____ and _____. We can conclude the first insight from the _____ and the second from the _____.

Question 04: Part (b)

[15 minutes] [Marks: 5]

A commercial bank wants to automate its loan approval process by predicting whether a loan applicant will default on repayment. The risk team has collected monthly income data (PKR thousands) from 8 past applicants and recorded whether each one defaulted (1 = defaulted, 0 = did not default). Using Logistic Regression, you will model the probability of default given an applicant's monthly income, then classify two new applicants to decide whether to approve or reject their loan.

Applicant	A1	A2	A3	A4	A5	A6	A7	A8
Monthly income X (PKR thousands)	20	30	40	50	60	70	80	90
Defaulted Y (1=yes, 0=no)	1	1	1	1	0	0	0	0

Intercept: $\beta_0 = +5.40$

Baseline log-odds when income = 0

Slope: $\beta_1 = -0.09$

Change in log-odds per PKR 1k income rise

- For Mr. Hamza ($X = 45$), and Ms. Nadia ($X = 70$) compute the log-odds z using $z = \beta_0 + \beta_1 \cdot X$. [1 Mark]
- Using the sigmoid formula $P = 1/(1 + e^{-z})$, compute $P(Y=1 | X=45)$ for Mr. Hamza and $P(Y=1 | X=70)$ for Ms. Nadia. [3 Marks]
- Apply the bank's decision rule to Mr. Hamza's and Ms. Nadia's probability using the 0.5 threshold. Should the bank approve or reject his/her loan? [1 Mark]

CLO # 4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas

Question 05:

[20 minutes]

[Marks: 2 + 5 + 2 + 2 + 4 = 15]

Consider building a decision tree, using the 7 examples in the table, to predict Read (yes/no).

The attributes and their values are:

- Fiction: yes/no
- Age: new/old/ancient
- Award: yes/no
- Rating: hi/med/low
- Length: short/medium/long

Ex.	Fiction	Age	Award	Rating	Length	Read
1	no	new	no	hi	short	Yes
2	yes	ancient	no	med	long	No
3	yes	old	yes	hi	medium	Yes
4	yes	old	no	med	medium	No
5	yes	ancient	yes	low	short	No
6	no	new	no	low	short	No
7	yes	new	no	hi	short	Yes

- Show step-by-step calculation of the entropy of the full dataset.
- Show step-by-step calculation of the information gain for each given attribute.
- Based on your calculations, which attribute is selected as the root node?
- Without referring to your calculated gain scores, explain qualitatively why the selected attribute is the best choice for the root node.
- Draw the final decision tree.

CLO # 4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas

Question 06: Part (a)

[10 minutes] [Marks: 5]

You are provided with the following dataset consisting of 10 observations. The goal is to predict Credit Approval (Y) based on Years of Experience (X_1), Savings in \$1000s (X_2), and Existing Debt (X_3).

ID	1	2	3	4	5	6	7	8	9	10
Experience (X_1)	2	1	5	8	1	10	3	2	4	6
Savings (X_2)	10	5	20	15	2	30	8	4	25	12
Existing Debt (X_3)	Low	High	Low	High	Low	High	High	Low	Low	High
Credit Approved (Y)	Yes	No	Yes	Yes	No	No	Yes	No	Yes	Yes

Consider a new applicant with the following profile: Experience = 4, Savings = 10, Debt = Low.

- Using Euclidean Distance on the numerical features (X_1 and X_2 only), calculate the distance between the new applicant and IDs 1, 7, and 8. (3 marks)
- If $K=3$ (considering all 10 data points and using X_1 and X_2 only), identify the three nearest neighbors and predict the class Y for the new applicant. (2 marks)

Question 06: Part (b)

[10 minutes] [Marks: 5]

A small clothing brand wants to understand how its monthly advertising spend (PKR thousands) affects its monthly sales revenue (PKR thousands). The marketing manager has collected data from the past 6 months. Using simple linear regression, you will find the best-fit line in the form $y = mx + b$, where x is advertising spend and y is predicted sales revenue, then use it to forecast sales for a planned future advertising budget.

Month	January	February	March	April	May	June	n = 6
Advertising spend X (PKR thousands)	10	20	30	40	50	60	$\Sigma X = ?$
Sales revenue Y (PKR thousands)	25	45	60	80	95	115	$\Sigma Y = ?$

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- Compute $\sum X$, $\sum Y$, the mean $\bar{x}$, and the mean $\bar{y}$ from the training data. Show your addition clearly. (1 mark)
- For each month, calculate $(x_i - \bar{x})$, $(y_i - \bar{y})$, the product $(x_i - \bar{x})(y_i - \bar{y})$, and $(x_i - \bar{x})^2$. Then sum the last two columns to get $\sum (x_i - \bar{x})(y_i - \bar{y})$ and $\sum (x_i - \bar{x})^2$. (2 marks)
- calculate the slope m and intercept b . (1 mark)
- Predict July's expected sales revenue by having $X = 75$. (1 mark)

Question 06: Part (c)

The following 6 data points are given in 2D space (X_1, X_2): Apply K-Means clustering with $K=2$. Use $P1=(1,1)$ as initial centroid C_1 and $P4=(8,7)$ as initial centroid C_2 . [10 minutes] [Marks: 2+2+1]

- Calculate the Euclidean distance of each point from both centroids in Iteration 1. Show all calculations. Assign each point to the nearest centroid.
- Compute the new centroids after Iteration 1. Then perform Iteration 2, reassign all points and check for convergence.
- State the final clusters and explain in one sentence why K-Means is sensitive to the choice of initial centroids.

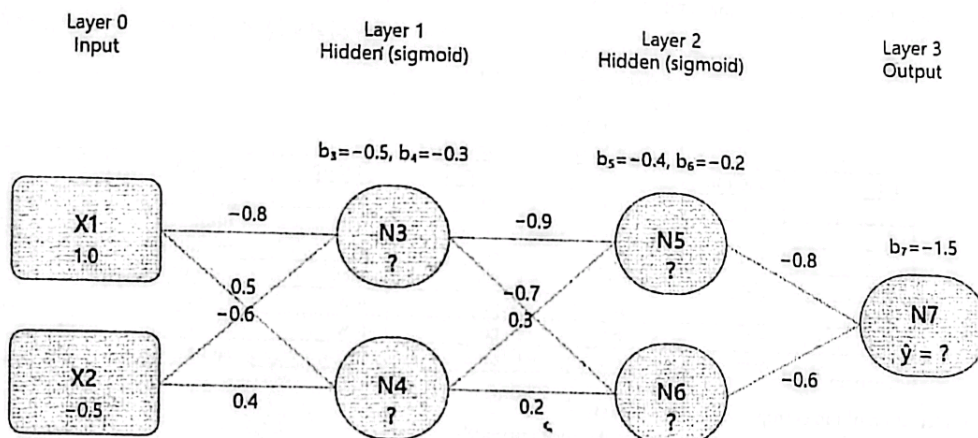
Point	P1	P2	P3	P4	P5	P6
X_1	1	1	2	8	9	8
X_2	1	3	2	7	8	9

CLO # 4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas

Question 07:

Part (a) Consider a multilayer feed-forward neural network given below. Calculate output at node N7. Show all steps for each calculation. Assume that sigmoid activation function is used in the network. (7.5 marks) [20 minutes] [Marks: 15]

Part (b) Also provide one iteration of Back Propagation after weight and bias updates by using back-propagation algorithm. Expected output for N7 is 0.5 and also learning rate is 0.5 (7.5 marks)



CLO # 3: Apply methods of blind as well as informed search on the problem related to the programs

Question 08

Using the grid world above, answer the following:

[15 minutes] [Marks: 10]

- Write the Q-learning update rule and identify the elements of its environment. [2 mark]
- An agent is in state (1,2) and takes action Up, moving to the Goal state (0,2) and receiving a reward of +100 and Learning rate is 0.5. Given the parameters below, write the full Q-learning equation with values substituted. [2 marks]
- Calculate the updated Q-value for state (1,2), action Up. Show all steps clearly. [3 marks]
- After this update, the agent is in state (0,1) and takes action Up, receiving reward +100. The max $Q(s', a')$ for the Goal state is now the value you just calculated in part (c). Using the same α and γ , calculate the new $Q(0,1, Up)$. [3 mark]

